



University
of Glasgow

Power Electronics

Introduction (Week 11 & 12)

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With Prof Lianping Hou

- Electric Circuits
- Power Switches
- Uncontrolled and Controllable Switches
- Heatsinks

With me (Prof Atif Jafri)

- Week-11
 - Switched Mode Power Supplies (DC-DC converters)
 - Snubber Circuits
- Week-12
 - DC-AC converters (Inverters)
 - Applications and Systems



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Power Electronics

Switched Mode DC-DC Converters I:

Concepts and Buck Converter

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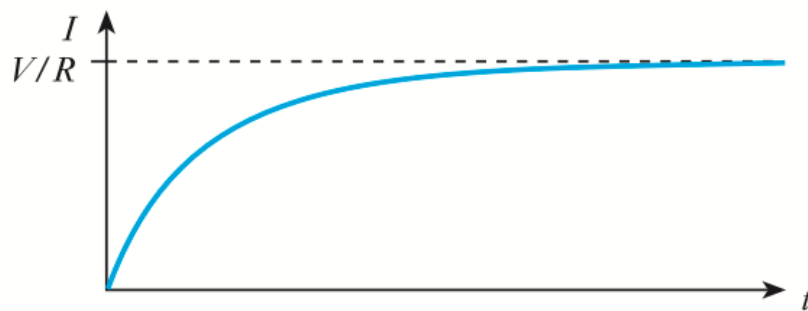
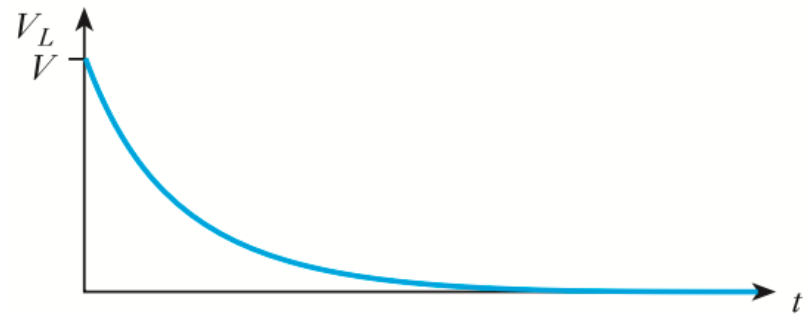
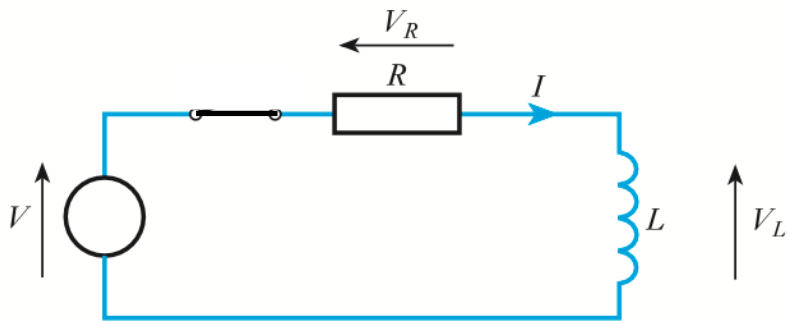
Please read **Chapter 7** in the textbook

POWER ELECTRONICS

Converters, Applications, and Design

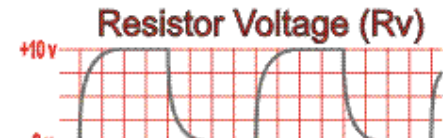
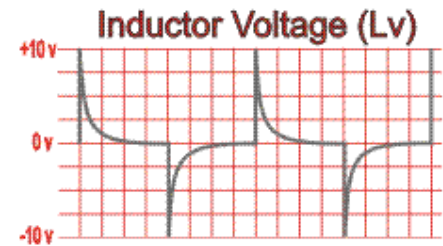
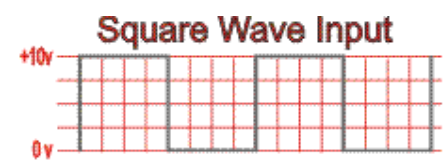
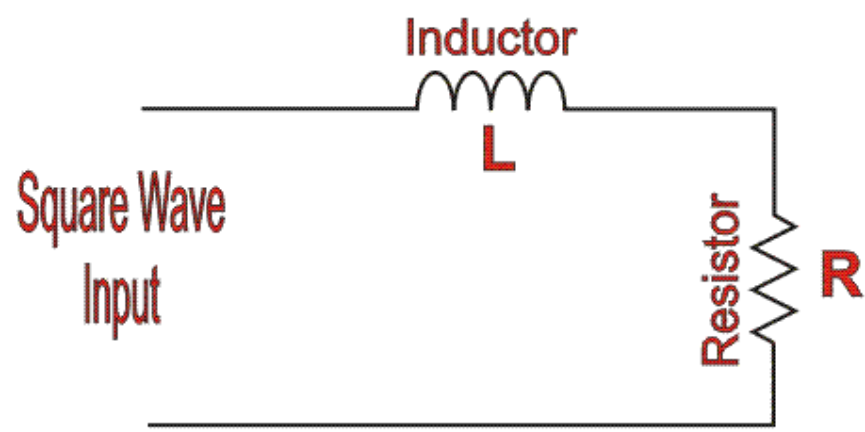
Background: RL Circuit

Recall the basic RL series circuit :



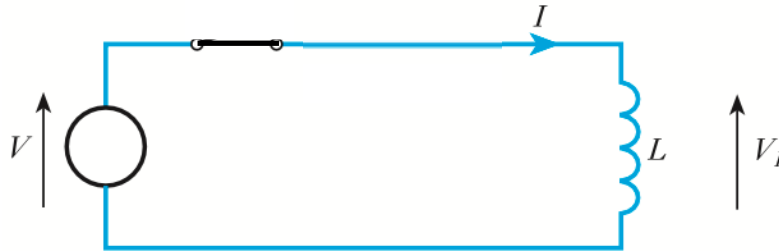
We can write the voltage equation for this circuit as

$$V = V_R + V_L = IR + L \frac{dI}{dt}$$



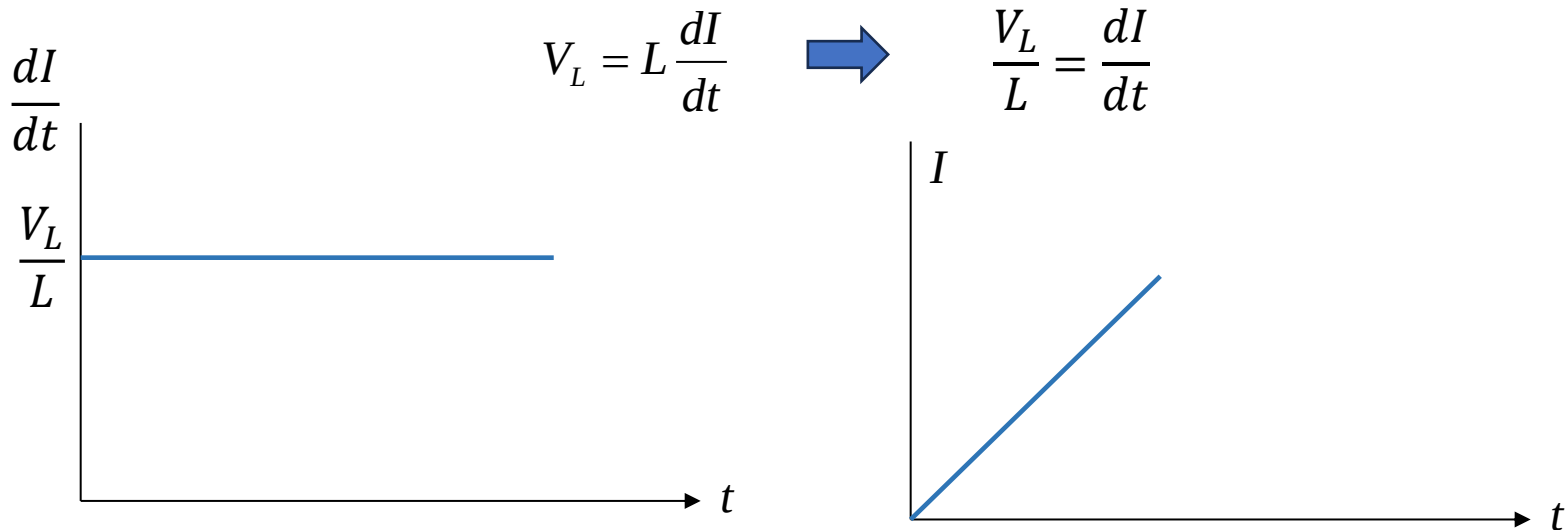
Background: RL Circuit

$$V = \cancel{V_R} + V_L = \cancel{IR} + L \frac{dI}{dt}$$



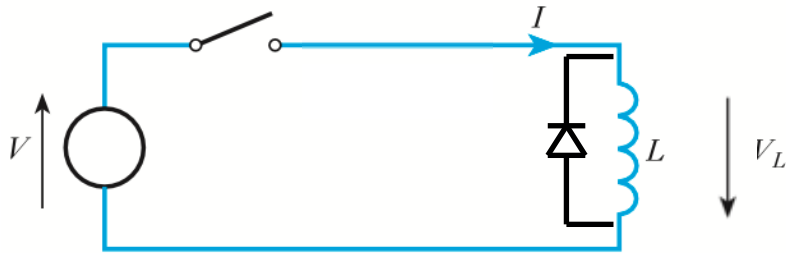
When the value of R is very small, we can neglect the " IR " term

Hence rate of change in current is constant.

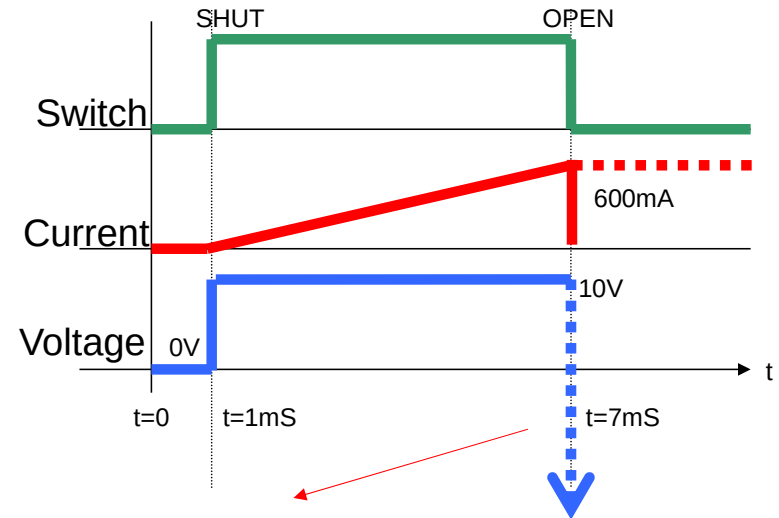


As the current in the inductor rises, energy is stored in the magnetic field.

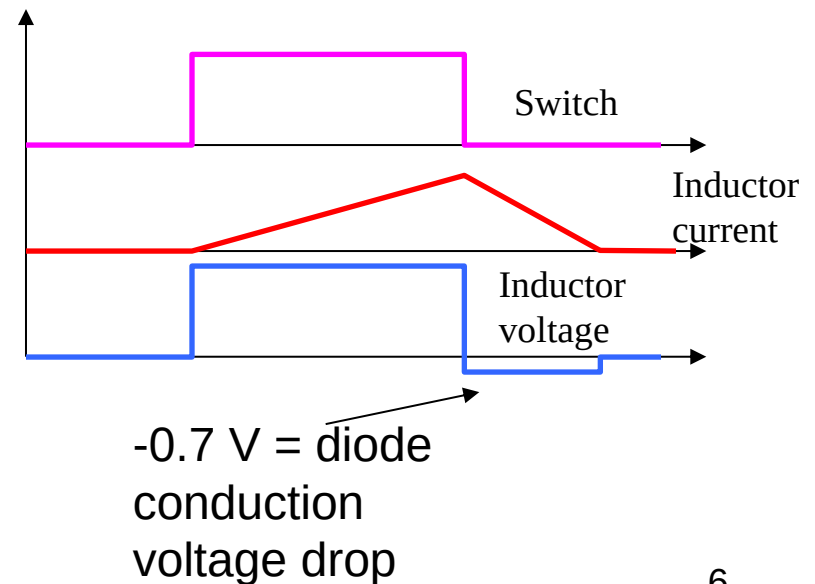
When the switch is opened, the inductor tries to maintain the current flowing through it and thus the voltage across the inductor reverses.



It is necessary to provide a return path for the inductor current when the switch is opened, and this is usually done by adding a diode across the inductor which is called as **freewheeling** or **flyback** diode.

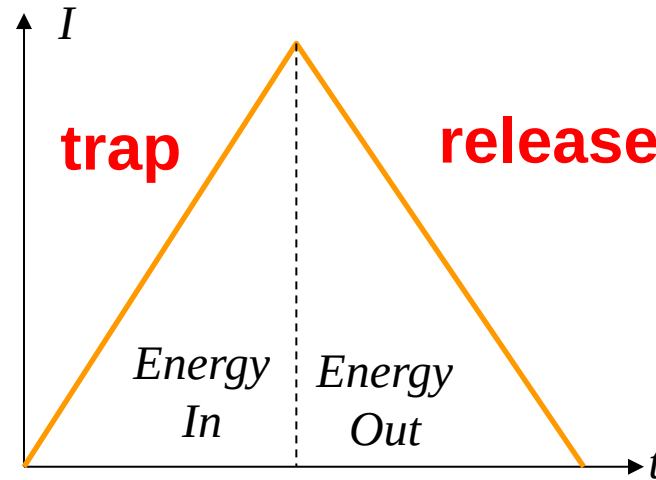


Sudden Switch Off will result in **Voltage Spike**, and **Large dv/dt**



Background: Energy in Inductor

The energy stored in an inductor is $\frac{1}{2}LI^2$. When current through the inductor is increasing, additional energy is being stored in the magnetic field. If the current is decreasing, then energy is being removed from the magnetic field. Assuming a perfect inductor, the energy (i.e., number of joules) returned to the circuit is exactly the same as was originally added.



For Power Conversion, released power from energy storage devices is expected to be delivered to the load as much as possible. Dissipative Energy Recovery is not acceptable.

We will see how to recover the trapped energy in switch-mode power conversion?

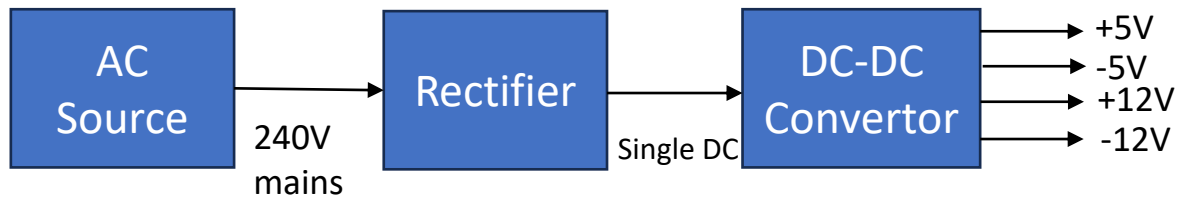
With this background knowledge, now we will explore following:

- Where we need DC – DC converters?
- Why switched mode DC – DC converter?
- What are different types of DC – DC converter?
 - How they work?
 - Expression for output voltage, average current, etc.
 - Important waveforms
 - Numerical Problems

Where we need DC – DC converters

In a mobile phone, battery voltage is converted into multiple different DC voltages for the microprocessor, display, **RF** amplifier and camera.

All PC power supplies provide at least +5VDC, +12VDC, and -12VDC. Some power supplies also provide +3.3VDC, -5VDC



Solution: DC-DC Converters

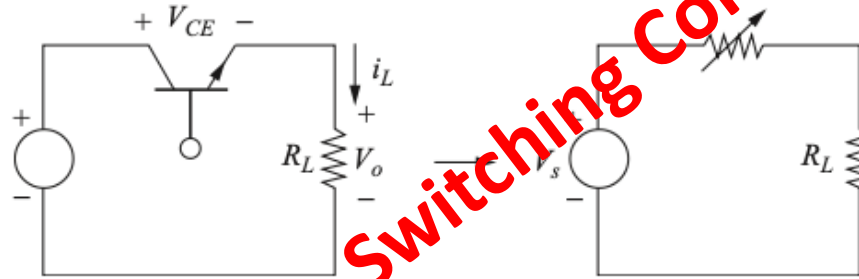
DC-DC converters are power electronic circuits that convert a dc voltage to a different dc voltage level, often providing a regulated output.

- Linear DC-DC Converters
- Switching DC-DC Converters or Switched Mode Power Supplies(SMPS)

Why switched mode DC – DC converter

Linear DC-DC Converters

By adjusting the transistor base current, the output voltage may be controlled over a range of 0 to roughly V_s .

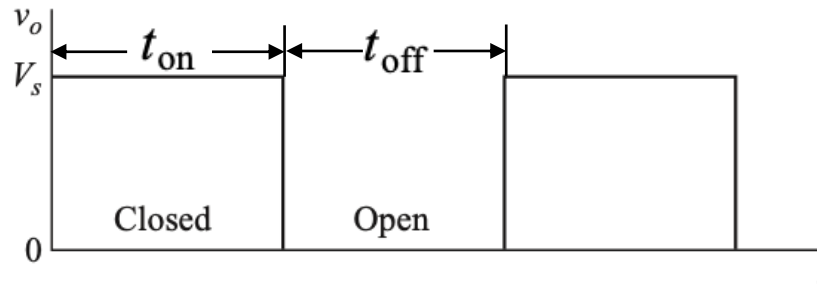
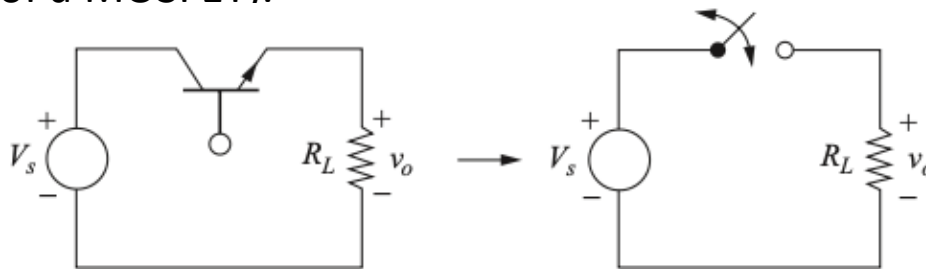


- This type of circuit is called a linear dc-dc converter or a linear regulator because the transistor operates in the linear region, rather than in the saturation or cutoff regions.
- Low efficiency of this circuit is a serious drawback for power applications.
 - Power absorbed by the load is $V_o I_L$
 - Power absorbed by the transistor is $V_{CE} I_L$
- If $V_o = 0.25 V_s$ then transistor will absorb 75% of power (low output voltage low efficiency)

Why switched mode DC – DC converter

Switching DC-DC Converters or Switched Mode Power Supplies(SMPS)

In a switching converter circuit, the transistor operates as an electronic switch by being completely **ON** or completely **OFF** (saturation or cutoff for a BJT or the triode and cutoff regions of a MOSFET).



Average Voltage

$$D = \frac{t_{on}}{t_{on} + t_{off}} = \frac{t_{on}}{T}$$

$$t_{on} = DT \text{ and } t_{off} = (1 - D)T$$

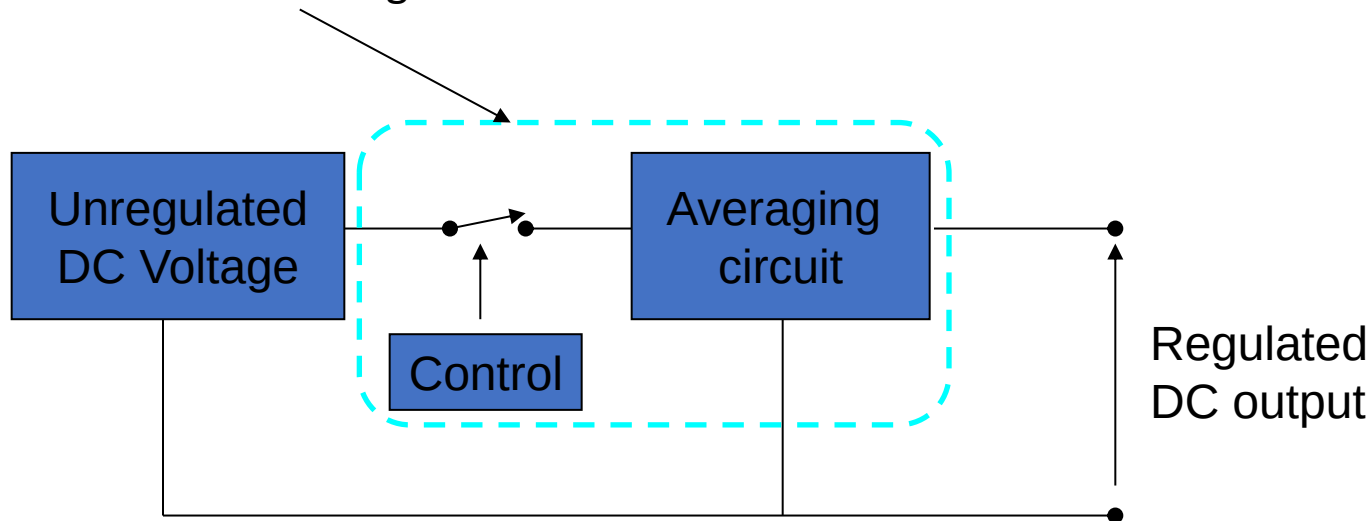
$$V_o = \frac{1}{T} \int_0^T v_o(t) dt = \frac{1}{T} \int_0^{DT} V_s dt = V_s D$$

What are different types of DC – DC converter

SMPSs come in many different varieties. For the purposes of this course, we will consider three of the most used types:

1. The step-down or “buck” converter. [$V_{out} < V_{in}$]
2. The step-up or “boost” converter. [$V_{out} > V_{in}$]
3. The “buck-boost” [$V_{out} <> V_{in}$]
4. The flyback converter. [$V_{out} <> V_{in}$]

The basic SMPS block diagram is as follows:

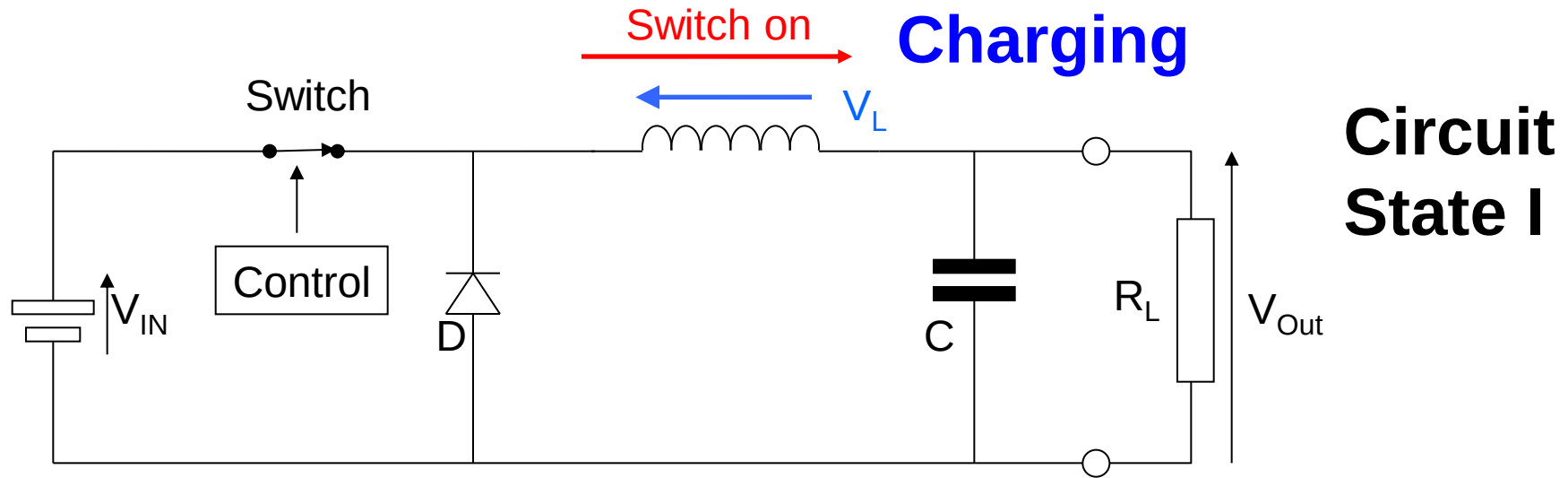


The two important components here are the switch (usually a MOSFET) and the averaging circuit that will include an inductor.

The Buck Converter

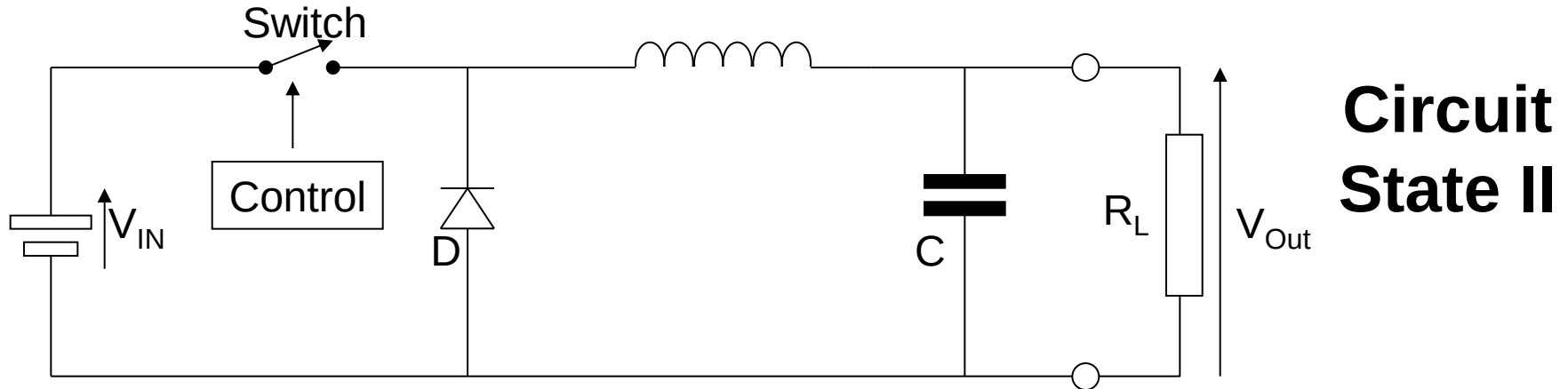
- How it works?
- Important waveforms
- Expression for output voltage
 - Based on average voltage of inductor
 - Based on change in current of inductor
- Numerical Problems

The Buck Converter



The circuit shows the SMPS circuit for a buck converter. When the switch is first closed, current starts to flow through the inductor and into the capacitor. The voltage across the inductor, V_L , is the difference between the input voltage V_{IN} and the output voltage V_{Out} . The diode is reverse biased 偏置 (“off”) by the supply voltage V_{IN} and does not conduct any current. The rate of change of current into the capacitor is limited by the inductor ($V = L \, di/dt$). At some point T_{off} the switch is opened. At this point the current through the inductor has risen to I_{Max} and the energy in the inductor ($\frac{1}{2}LI^2$) is stored in its magnetic field.

Discharging

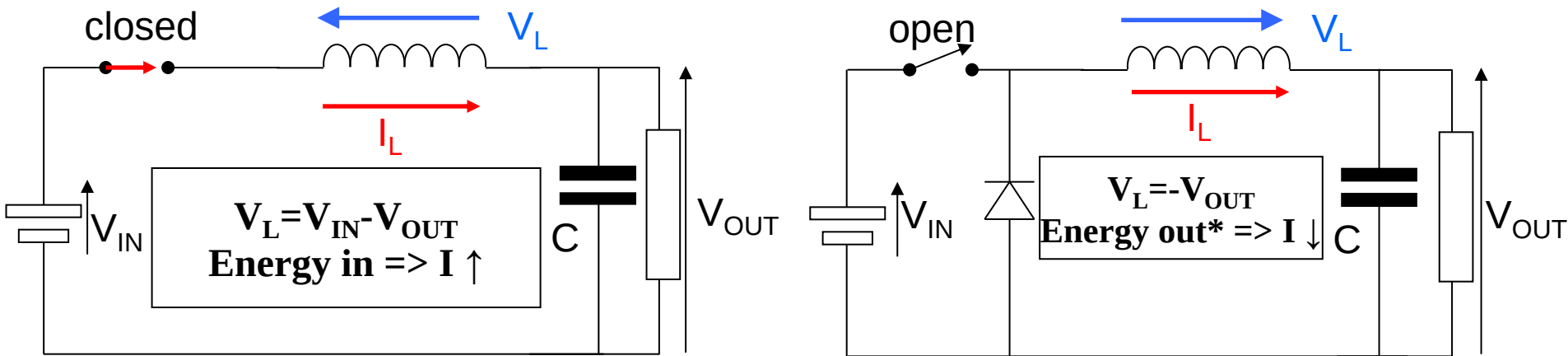
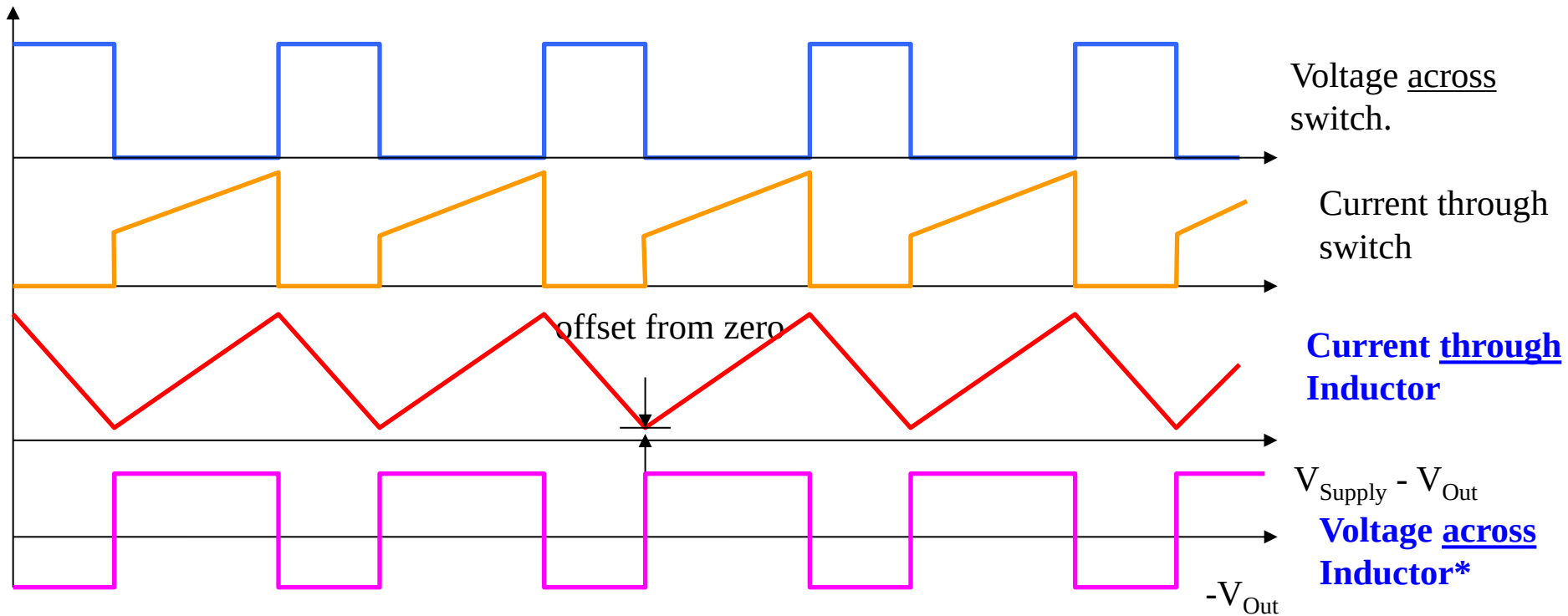


As soon as the switch is opened, the potential on the switched side of the inductor goes sharply negative (but the current continues to flow) as the energy in the magnetic field tries to maintain the current through the inductor. This has the effect of bringing the diode into conduction (hence $V_L = -V_{Out} + 1V$), thereby completing the circuit and enabling the current to continue to flow into the capacitor. As the energy in the inductor falls so the current diminishes but the capacitor continues to be charged (more slowly). If we ignore the diode voltage drop, the voltage across the inductor is now $-V_{Out}$.

The switching frequency, supply voltage and inductor value are all chosen such that there is always current flowing in the inductor. This is called continuous mode, i.e., continuous current in the inductor.

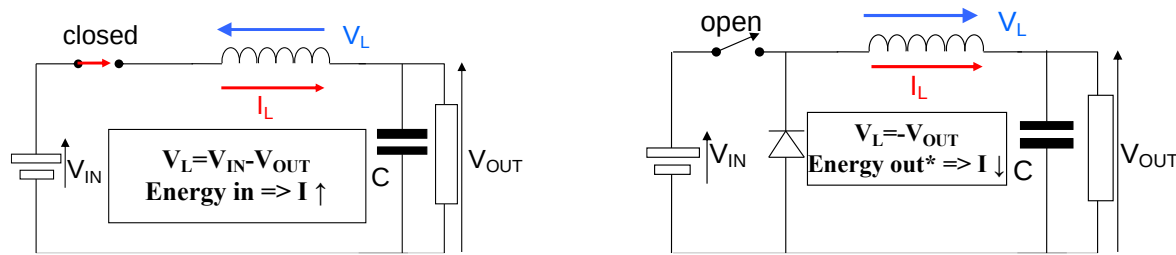
All of the circuitry necessary to implement the control function is available in an IC.

Buck converter switching waveforms.



* Ignoring the voltage drop across the diode, which we will do to simplify the analysis.

Buck converter Input/Output Voltage Relation



The integral of inductor voltage is equal to the change in the inductor flux linkage.
The net change of flux linking the inductor over one time period of repetition is zero, which is a necessary condition for steady-state operation

$$\int_0^T V_L dt = 0$$

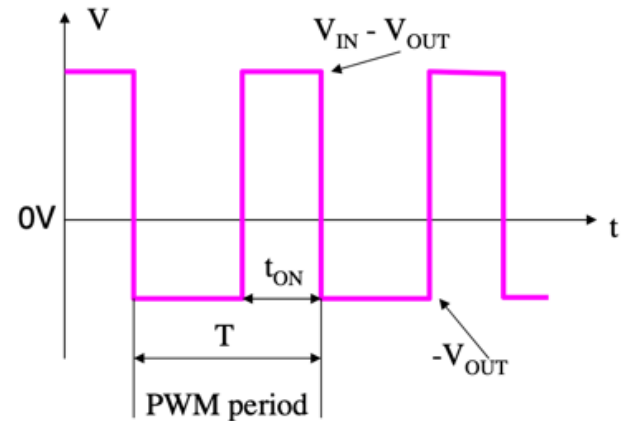
$$\int_0^{t_{on}} (V_{IN} - V_{OUT}) dt + \int_{t_{on}}^T -V_{OUT} dt = 0$$

$$(V_{IN} - V_{OUT}) t_{on} - V_{OUT}T + V_{OUT}t_{on} = 0$$

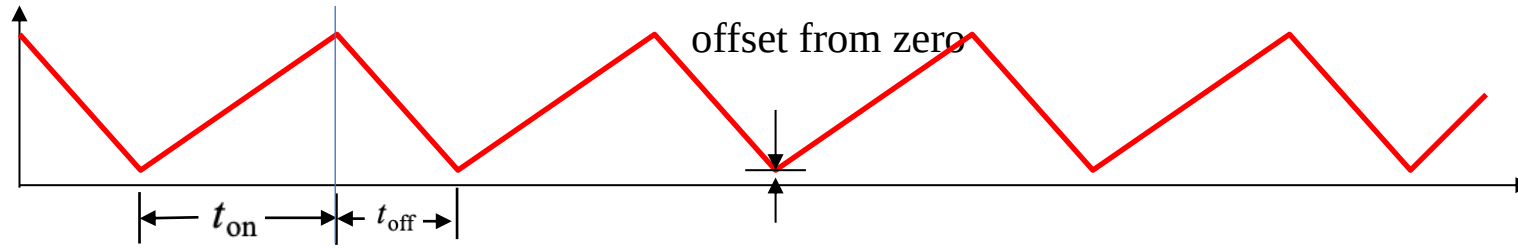
$$V_{IN}t_{on} - V_{OUT}t_{on} - V_{OUT}T + V_{OUT}t_{on} = 0$$

$$V_{IN}t_{on} = V_{OUT}T \quad \Rightarrow \quad \frac{t_{on}}{T} = \frac{V_{OUT}}{V_{IN}}$$

$$D = \frac{V_{OUT}}{V_{IN}} \quad \Rightarrow \quad \mathbf{V_{OUT} = DV_{IN}}$$



Buck converter inductor current change



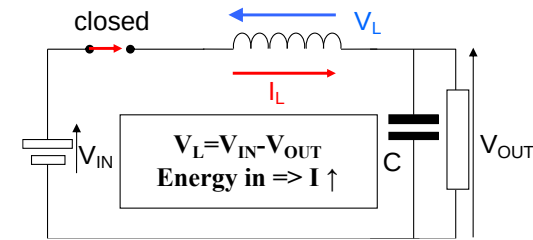
Current through inductor

Computing the change in inductor current using following expression:

$$V_L = L \frac{di_L}{dt} \Rightarrow \frac{di_L}{dt} = \frac{V_L}{L}$$

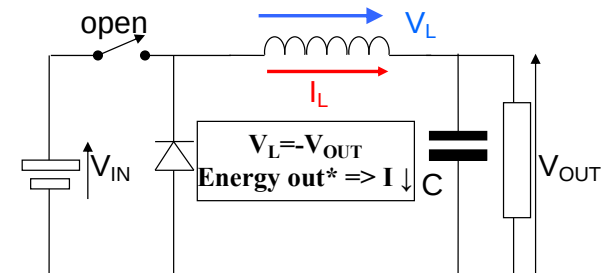
When switch is closed:

$$\frac{(\Delta I_L)_{\text{switch closed}}}{t_{on}} = \frac{V_{IN} - V_{OUT}}{L} \Rightarrow (\Delta I_L)_{\text{switch closed}} = \frac{V_{IN} - V_{OUT}}{L} \times t_{on}$$

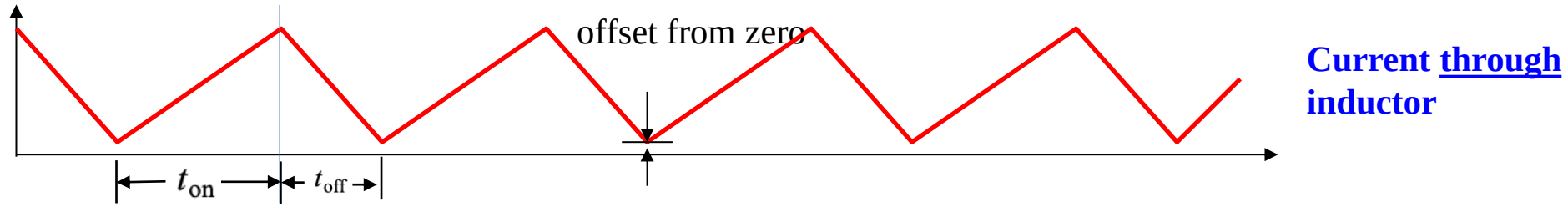


When switch is open:

$$\frac{(\Delta I_L)_{\text{switch open}}}{T - t_{on}} = \frac{-V_{OUT}}{L} \Rightarrow (\Delta I_L)_{\text{switch open}} = \frac{-V_{OUT}}{L} \times (T - t_{on})$$



Buck converter Input/Output Voltage Relation using change in inductor current



We can prove input ~ output voltage relation by using Net change in inductor current is zero over one period

$$(\Delta I_L)_{switch\ closed} + (\Delta I_L)_{switch\ open} = 0.$$

$$\frac{V_{IN} - V_{OUT}}{L} \times t_{on} + \frac{-V_{OUT}}{L} \times (T - t_{on}) = 0$$

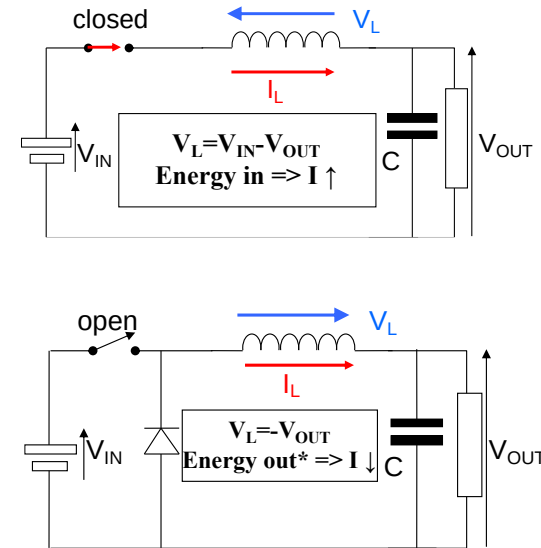
$$\frac{V_{IN} - V_{OUT}}{L} \times t_{on} = \frac{V_{OUT}}{L} \times (T - t_{on})$$

$$\frac{V_{IN} - V_{OUT}}{V_{OUT}} = \frac{(T - t_{on})}{t_{on}}$$

$$\frac{V_{IN}}{V_{OUT}} - 1 = \frac{T}{t_{on}} - 1$$

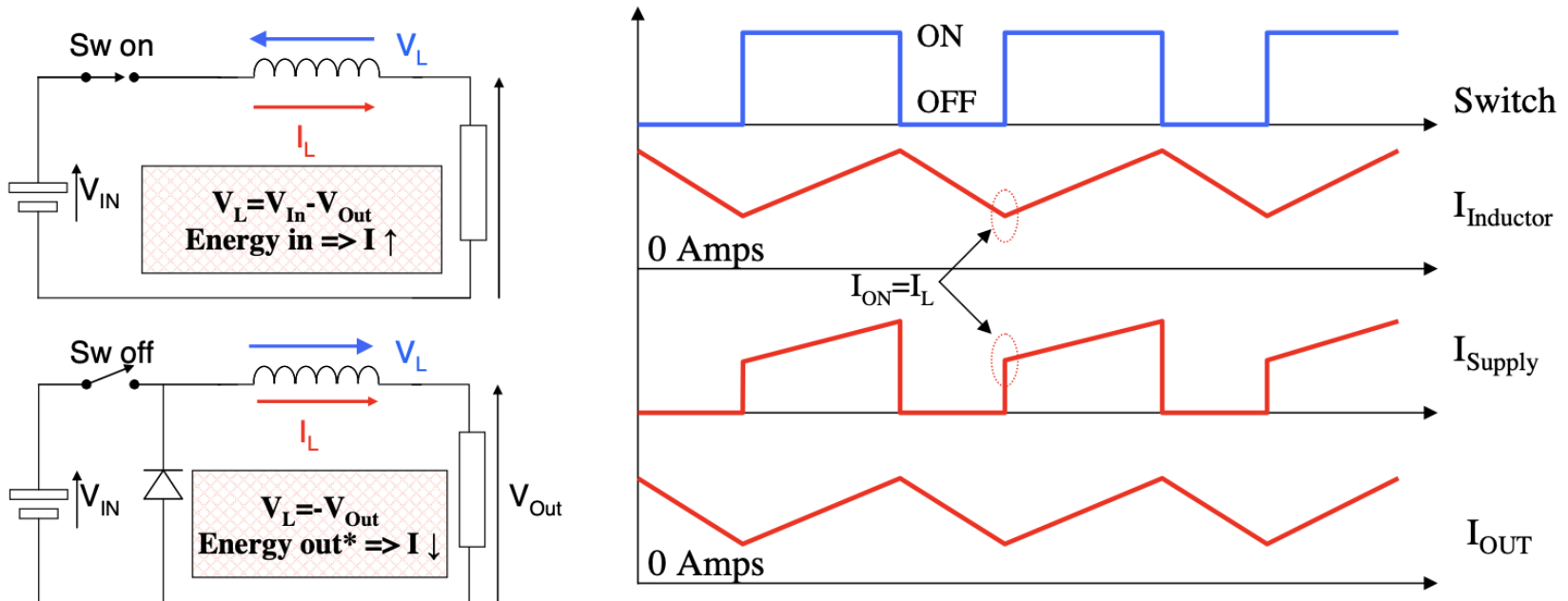
$$\frac{V_{IN}}{V_{OUT}} = \frac{T}{t_{on}} \Rightarrow \frac{V_{OUT}}{V_{IN}} = \frac{t_{on}}{T} \Rightarrow \frac{V_{OUT}}{V_{IN}} = D$$

$$V_{OUT} = DV_{IN}$$



Input ~ Output Power (Loss less system)

Reminding ourselves about the currents in the circuit:



Sticking with our assumption of a lossless system:

$$\text{Output power} = \text{input power} \Rightarrow V_{OUT} I_{OUT(\text{Mean})} = V_{IN} I_{IN(\text{Mean})}$$

$$\text{Setting } V_{OUT} = D V_{IN}, \text{ we have } I_{OUT(\text{Mean})} = I_{IN(\text{Mean})} / D$$

(Remember for a DC voltage, average power = $V_{DC} \times I_{\text{Mean}}$)

Minimum and Maximum Inductor Current

$$I_{L,avg} = \frac{V_{OUT}}{R} \text{ or } \frac{Power_{avg,OUT}}{V_{OUT}}$$

$$I_{L,max} = I_{L,avg} + \frac{(\Delta i_L)_{switch\ closed}}{2}$$

$$I_{L,min} = I_{L,avg} - \frac{(\Delta i_L)_{switch\ closed}}{2}$$

Note: Since change in inductor current is zero, you can take $(\Delta i_L)_{switch\ closed}$ or $(\Delta i_L)_{switch\ open}$ to calculate max and min current. You need to consider the sign of the current.

Buck converter Output Voltage Ripples

- So far, the capacitor was assumed to be very large to keep the output voltage constant.
- In practice, the output voltage cannot be kept perfectly constant with a finite capacitance.
- The variation in output voltage, **or ripple**, is computed from the voltage-current relationship of the capacitor.

$$i_C = i_L - i_R$$

$$\Delta Q = C \Delta V_{OUT} \quad \Delta V_{OUT} = \frac{\Delta Q}{C}$$

$$\Delta V_{OUT} = \frac{1}{2} \times \frac{T}{2} \times \frac{\Delta I_L}{2} \times \frac{1}{C}$$

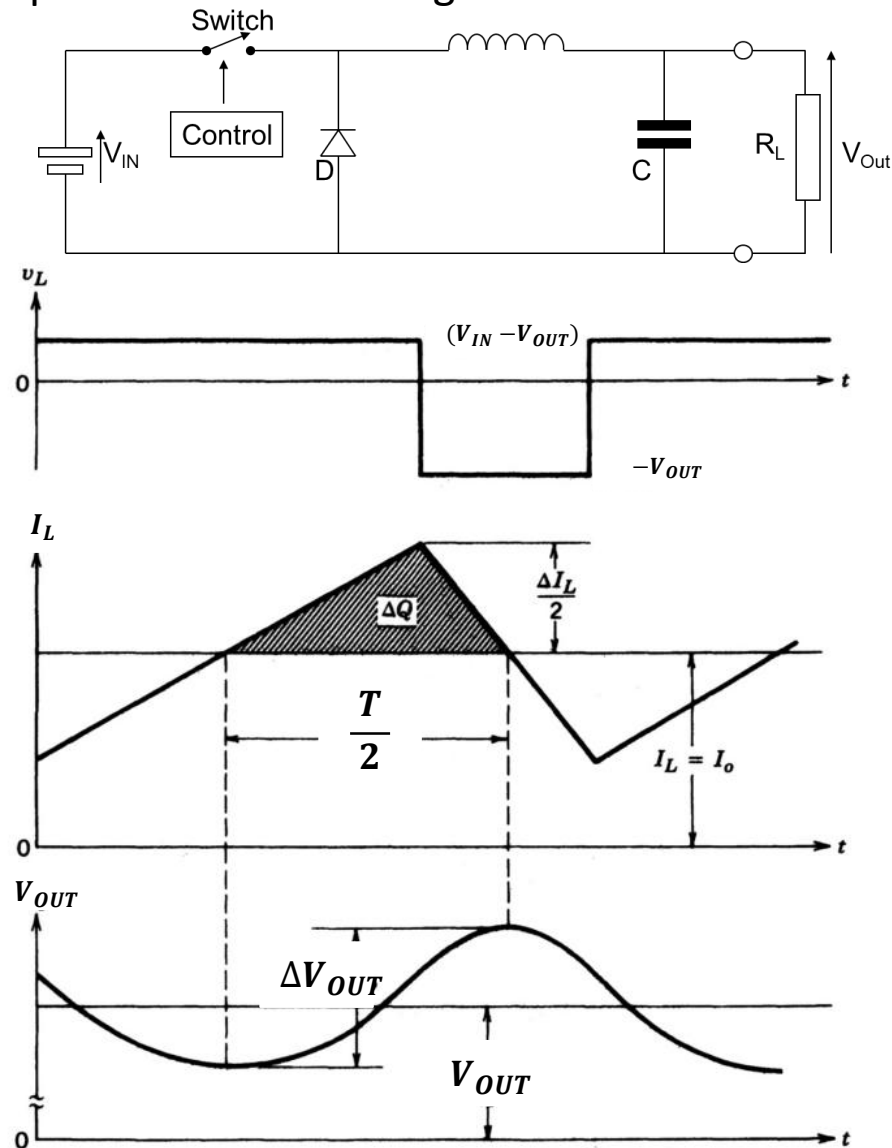
$$\Delta I_L = \frac{V_{OUT}}{L} \times (T - t_{on})$$

From Previous slide

$$\Delta V_{OUT} = \frac{1}{8} \times \frac{T}{C} \times \frac{V_{OUT}}{L} \times (T - t_{on})$$

$$T - t_{on} = T(1 - D)$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{T^2}{8CL} \times (1 - D) = \frac{(1 - D)}{8CLf^2}$$



Buck tutorial.

A buck converter has following parameters. Determine output voltage, average, minimum and maximum inductor current and output voltage ripple.

$$V_{IN} = 50 \text{ V}$$

$$D = 0.4$$

$$L = 400 \text{ } \mu\text{H}$$

$$C = 100 \text{ } \mu\text{F}$$

$$f = 20 \text{ kHz}$$

$$R = 20 \text{ } \Omega$$

$$V_{OUT} = DV_{IN}$$

$$= (50)(0.4) = 20 \text{ V}$$

$$I_{L,avg} = \frac{V_{OUT}}{R}$$

$$= \frac{20}{20} = 1 \text{ A}$$

Buck tutorial.

A buck converter has following parameters. Determine output voltage, average minimum and maximum inductor current and output voltage ripple.

$$V_{IN} = 50 \text{ V}$$

$$D = 0.4$$

$$L = 400 \text{ } \mu\text{H}$$

$$C = 100 \text{ } \mu\text{F}$$

$$f = 20 \text{ kHz}$$

$$R = 20 \text{ } \Omega$$

$$(\Delta I_L)_{\text{switch closed}} = \frac{V_{IN} - V_{OUT}}{L} \times t_{on}$$

$$= \frac{50 - 20}{400 \mu\text{H}} \times 20 \mu\text{s}$$

$$= 1.5 \text{ A}$$

$$I_{L,max} = I_{L,avg} + \frac{(\Delta i_L)_{\text{switch closed}}}{2}$$

$$= 1 + \frac{1.5}{2} = 1.75 \text{ A}$$

$$T = \frac{1}{f} \text{ and } t_{on} = DT$$

$$T = \frac{1}{20 \text{ kHz}} = 50 \mu\text{s}$$

$$t_{on} = 0.4 \times 50 = 20 \mu\text{s}$$

$$I_{L,min} = I_{L,avg} - \frac{(\Delta i_L)_{\text{switch closed}}}{2}$$

$$= 1 - \frac{1.5}{2} = 0.25 \text{ A}$$

Buck tutorial.

A buck converter has following parameters. Determine output voltage, average minimum and maximum inductor current and output voltage ripple.

$$V_{IN} = 50 \text{ V}$$

$$D = 0.4$$

$$L = 400 \text{ } \mu\text{H}$$

$$C = 100 \text{ } \mu\text{F}$$

$$f = 20 \text{ kHz}$$

$$R = 20 \text{ } \Omega$$

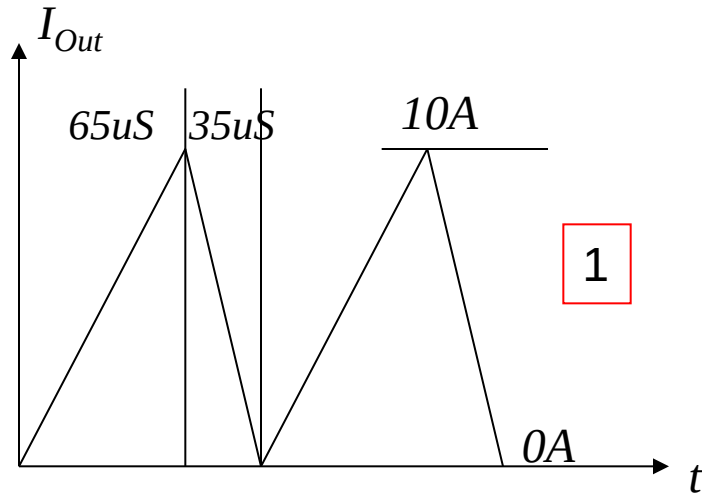
Output voltage ripple:

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{(1 - D)}{8CLf^2}$$

$$\begin{aligned} &= \frac{1 - 0.4}{8(400)(10)^{-6}(100)(10)^{-6}(20,000)^2} \\ &= 0.00469 = 0.469\% \end{aligned}$$

Buck tutorial.

Determine the mean output voltage, current and power for the following operating points. Assume 100% conversion efficiency and continuous mode operation. $V_{in}=100V$



1.

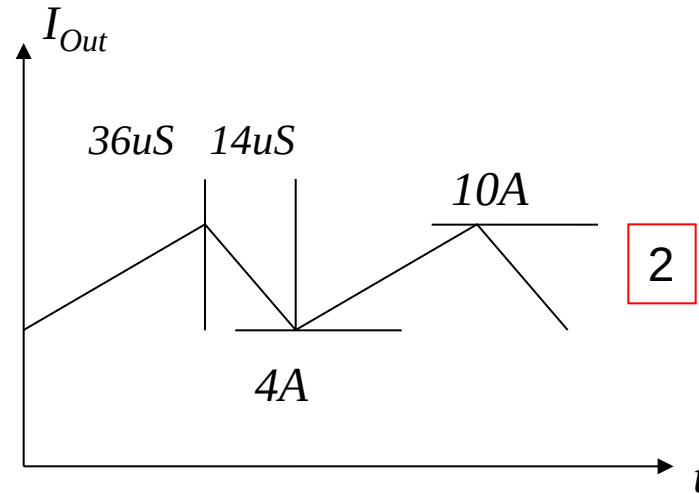
$$V_{in} = 100V$$

$$D = 0.65$$

$$V_{Out} = D \cdot V_{in} = 65V$$

$$I_{Out(Ave)} = (0.5 \times 65\mu s \times 10 + 0.5 \times 35\mu s \times 10) / (65\mu s + 35\mu s) \\ = 5A$$

$$P_{Out} = 325W$$



2.

$$V_{in} = 100V$$

$$D = 0.72$$

$$V_{Out} = D \cdot V_{in} = 72V$$

$$I_{Out(Ave)} = 7A$$

$$P_{Out} = 504W$$

Switched-mode Power Supplies (SMPS)开关电源

We saw how the linear power supply was very easy to build but suffered from relatively low efficiency. Where efficiency is an important consideration, the SMPS should be the supply of choice. For even moderately-sized power supplies the **advantages** of the SMPS efficiency can be considerable:

- 1.Heatsinks 散热片 can be smaller;
- 2.A cooling fan is not usually needed;
- 3.The power density 功率密度(watts per cm³) is greater.
- 4.The weight is lower.

However, against these advantages, the following **disadvantages** must be considered:

1. The SMPS is electrically very noisy.